



CODE SPORT

In this month's column, we take a short break from our discussion of compiler optimisations and feature a set of questions for computer science interviews.

We have been discussing compiler optimisation over the last few columns. However, a couple of our readers reminded me that we used to feature a set of programming questions for our December column—and this time, I missed doing that. Thank you, Sunil and Shreya, for the timely reminder. Hence, in this month's column, we will take a break from our discussion on inter-procedural optimisations and turn to discussing a few interview questions. Let's start off with a few puzzles, followed by programming questions.

This month's questions

1. Let A be a multi-set of integers consisting of N numbers. You are allowed to pick up two integers at a time. If both are even, or both odd, discard both and insert an even integer. If not, you discard the even integer. Given that there are an even number of odd integers initially in A , can you tell whether the last integer will be even or odd. What is the reasoning behind your answer?
2. There is a building with N floors. You are given M balls. You are told that there is an integer $K < N$ such that if the ball is dropped from the floor K or any floor higher than K , it would break. But the ball would not break if you drop it from a floor below K . What is the minimum number of balls that you need to drop to determine K : (a) in the best-case scenario; (b) in the worst-case scenario?
3. You have 100 doors in a row, all of which are initially closed. You make 100 passes over these doors, starting with the first door. The first time when you make the pass, you visit every door and toggle the door (i.e., if the door is open, close it; if it is closed, open it). During the second pass, you visit every second door (doors numbered 2, 4, 6, 8...and so on). During the third pass, you visit only every third door (doors marked 3, 6, 9, 12...). You repeat these passes until you finish all

the 100 passes. Now can you determine what state each door is in after the 100th pass? Which doors are open and which are closed?

4. You are given a coin and you have been told to use this for a coin-tossing experiment. You are told that the coin is weighted to come up heads more times than tails. Given such a biased coin, how can you get a fair coin toss?
5. You are given a sequence of numbers from 1 to $n-1$, with two of the numbers repeating twice. For instance, instead of $\langle 1, 2, 3, 4, 5 \rangle$, you are given $\langle 1, 2, 2, 3, 3, 4, 5 \rangle$. You need to find out the two numbers which are repeated. Can you find the two numbers if you are told that you can only use constant-size extra storage?
6. Three co-workers are discussing salary issues in their company. They want to know whether the average salary of an employee in their company is greater than the industry average. They have found the industry average salary from a magazine survey. For the average salary in their company, they decide to take the average of their salaries (they assume that they are the only three employees in the company). However they do not want to disclose their salaries. How can they take the average without disclosing their salaries?
7. You are travelling on a road and have reached a junction where two paths diverge. One path will take you to your correct destination. However you do not know which is the correct path, and alas, there are no sign boards! However, a truth teller and a liar are standing at the junction to help/confuse you. The truth-teller will always answer any question you ask truthfully. The liar will always answer any question you ask falsely. Now, you are allowed to ask only one question. You do not know who the truth-teller is and who the liar is. What is the single question you can ask to help you find the correct road to your destination?